

Math 120 Spring 2020 Worksheet 9

1. Your professor grades exams by choosing a random number between 0 and 100 and assigning that as the score. What is the probability that your score is greater than or equal to 80? Use the uniform distribution.

2. In a bank, on the average, a customer comes in every 10 minutes. Use the exponential distribution.

- (a) What is the probability that the next customer will come in 3 minutes?
- (b) What is the probability that the next customer will come between 5 and 15 minutes from now?
- (c) What is the probability that no customer will come in 8 minutes?

3. The scores on a certain standardized test are normally distributed with average $\mu = 500$ and standard deviation $\sigma = 100$. What is the probability that a randomly chosen student scores between 400 and 650 on the exam?

z	-2.0	-1.5	-1.0	-0.5	0	0.5	1.0	1.5	2.0
$\Phi(z)$	0.0228	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772